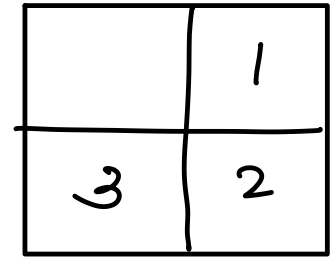


1234

(Start)

x

?



4132

(Goal)

y

Check : $I(x) = I(y)$?

$$\chi(x) = \begin{cases} +1 & \text{if blank is in 1 or 4}^{th} \text{ position} \\ -1 & \text{otherwise} \end{cases}$$

$$\text{Sgn}(x) = (-1)^{\text{Total no of inversions}}$$

$$\text{sgn}(4132) = (-1)^{\text{Total inversions in } 4132}$$

$$= (-1)^{3+0+1+0}$$

$$= (-1)^4$$

$$= +1$$

hash rank: (Sl. no in Lexicographic order)

$$\overset{c_1}{3}(3!) + \overset{c_2}{0}(2!) + \overset{c_3}{1}(1!) + \overset{c_4}{0}(0!)$$

$$= 19 = h(4132)$$

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$$I(y) = \operatorname{sgn}(y) \cdot \chi(y)$$

$$= (+1) \cdot (+1)$$

$$= \underline{\underline{+1}}$$

$$I(x) = \operatorname{sgn}(x) \cdot \chi(x)$$

$$(+1) \cdot (+1)$$

$$= \underline{\underline{+1}}$$

$h(1234) = 0.$

$(-1)^0 = \underline{\underline{+1}}$

Since $I(x) = I(y) = +1$,

we can reach y from x.

Result: $I(x) = \text{sgn}(x) \cdot \chi(x)$ is invariant for the n-puzzle.

Proof:

Take a legal move from x to y.

Then $I(y) = \text{sgn}(y)\,\chi(y)$

$$= (-\text{sgn}(x))(-\chi(x))$$

$$= \text{sgn}(x)\,\chi(x)$$

(Each legal move $x \to y$ changes position of blank from + to − or − to +) — transposition

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$= I(x)$

(similar for sgn)

Any permutation is a composition of transpositions.

$\Rightarrow$ The invariant holds for a set of legal moves

$\Rightarrow$ $\text{sgn}(x) \cdot \chi(x)$ is an invariant of n-puzzle.

Lexicographically ordered BFS:

(4-puzzle 4!=24)

Initialization:

Set visited $[0, \ldots, 23] \leftarrow$ False

distance $[v] \leftarrow \infty$ for all v

Parent $[v] \leftarrow$ Nil

Move $[v] \leftarrow$ Nil

$Q = \phi$

visited [R(start)] ← True

distance [start] ← 0

Enqueue (Q, start)

while $Q \neq \phi$

$m \leftarrow$ no. of states currently in Q

$N \leftarrow \phi$ (temporary list)

Repeat in times

$x \leftarrow$ Dequeue (Q)

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Generate every legal neighbour y of x (using L, U, R, D)

$r = h(y)$

updating records

If visited[r] is False

visited[r] ← True

distance[y] ← distance[x] + 1

Parent[y] ← x

move[y] ← legal move from x to y

Add y to N

Sort N according to Factorial Rank

For each y in N, Enqueue (Q, y).

Queue (FIFO)

Front → $[u_1, u_2, \ldots, u_n]$ ← Rear

↓

Dequeue (Q)

Removes front item

Enqueue (Q, x)

↓

Append x at rear.

Q: Implement Lexicographically BFS on 4-puzzle with start state as 1234.

A:

Q  empty

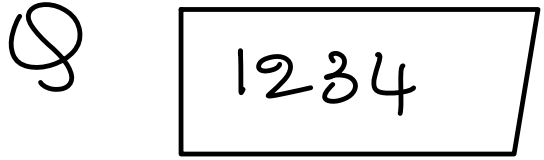
Start $\leftarrow$ 1234

visited $\left[h(\underset{1234}{\text{start}})\right]$ = visited (0) $\leftarrow$ True

distance [1234] $\leftarrow$ 0.

Enqueue (Q, 1234)

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$Q \neq \phi$

$m \leftarrow 1$

$N \leftarrow \phi$

Level 1 implementation of Lexicographic BFS.:

Repeat 1 time.

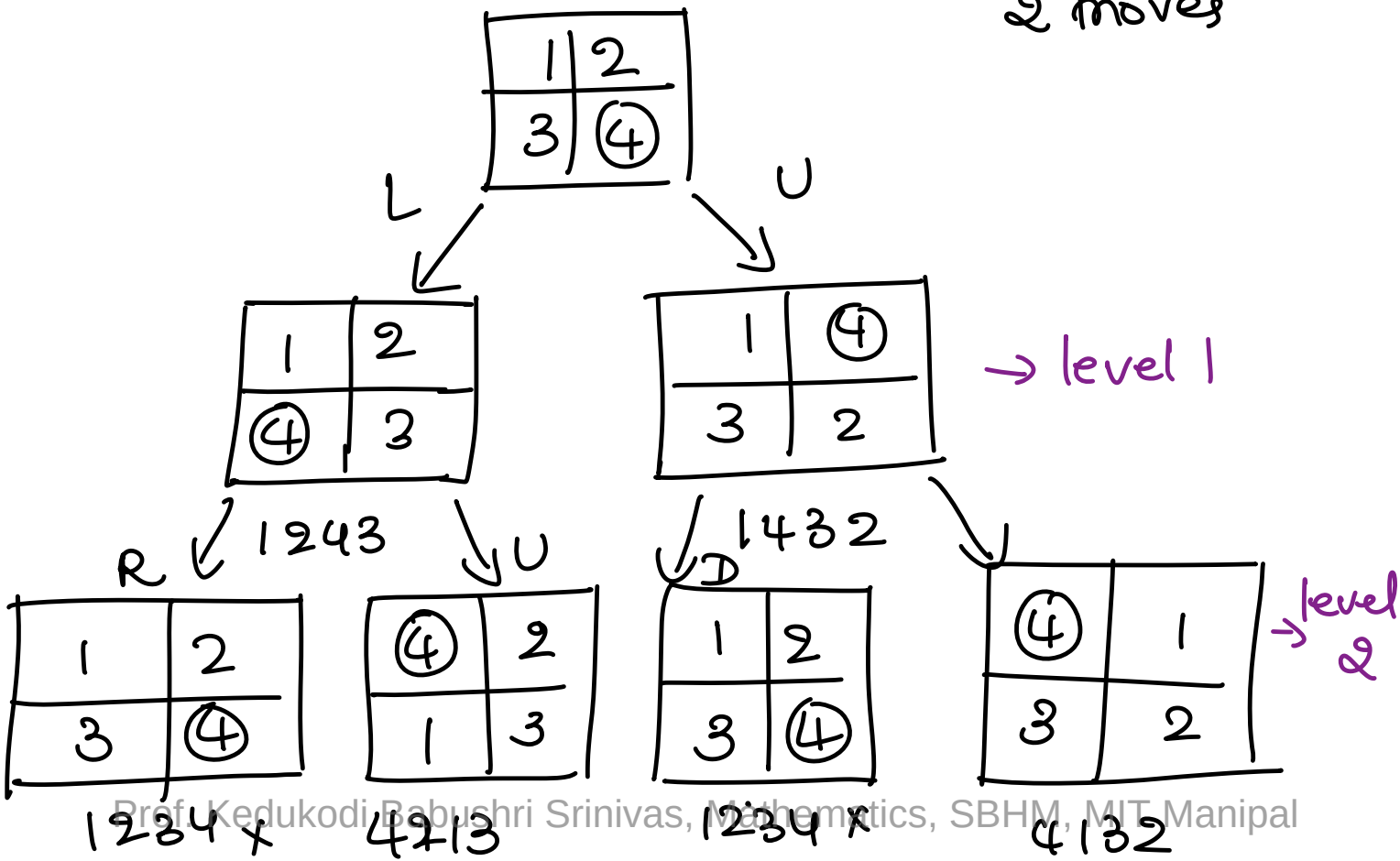
$x = \text{Dequeu}(Q)$

$= 1234$

(Removes 1234 from Q)

For every legal neighbour y of 1234

→ 2 legal moves



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$y = 1243$

3! 2! 1! 0!
0+0+1+0

$r = h(y) = h(1243) = 1.$

If visited[r] is False (Yes.)

visited[1] ← True (update)

distance(y) ← distance(x)+1

distance(1243) ← distance(1234) + 1
= 0+1 = 1.

parent(1243) ← 1234

move(1243) ← 'L'.

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$N = \{1243\}$ updated N.

$y = 1432.$

$3!\ 2!\ 1!\ 0$
$0+2+1+0$
$4+1=5$

$a = h(1432) = 5$

If visited[5] is False (Yes Update)

visited[5] $\leftarrow$ True

distance(1432) $\leftarrow 0+1 = 1$

Parent(1432) $\leftarrow$ 1234

Move(1432) $\leftarrow$ 'U'

$N = \{1243, 1432\}$ (rank 1, rank 5)

Using rank of lexicograph. order.

Sort N

$N = \{1243, 1432\}$ (Already sorted)

For y in N, Enqueue (Q, y)

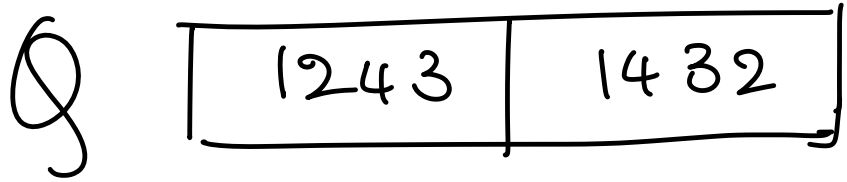
Enqueue $(Q, 1243)$

(Add from rear.)

Q | 1243 |

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Enqueue (Q, 1432)



Level 2 implementation :

$m \leftarrow 2$

$N \leftarrow \phi$

Repeat $2^{(m)}$ times.

x = Dequeue (Q) (Removes front item)

Q = {1243, 1432}

$x = 1243$

Every legal move y from 1243.

$y = 1234$

$r = R(y) = 1$

visited [1] . is False No. (it's true already visited)

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$y = 4213$

$x = h(4213) = 20$

Visited [20] is False Yes

visited [20] ← True

distance [4213] ← distance[1243] + 1

= 1 + 1

= 2

parent [4213] = 1243

move [4213] ← 'U'

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$N = \{4213\}$

(2nd time)

$x = 1432$

$y = 1234.$

visited[1] is False No

$y = 4132$

$r = h(4132) = 19$

visited[19] ← True

distance(4132) ← 1 + 1 = 2

Parent (4132) = 1243

move (4132) ← 'L'

$N = \{$ 4213 (rank 20), 4132 (rank 19) $\}$

(Lexicographic sorting)

↓ sort

$N = \{ 4132, 4213 \}$

Enqueue (0, 4132)

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Q | 4132 |

Enqueue (Q, 4213)

Q | 4132 | 4213 |

(Level 2 ended)

Distance & Level	States in processing order Q	hash ranks (RCS)
0	[1234]	0
1	[1243, 1432]	1, 5
2	[4132, 4213]	19, 20
3	[2413, 3142]	10, 13
4	[2314, 3124]	8, 12

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5	[2341, 3421]	9, 17
6	[4321]	23